

Review

Recent Advances in Pharmaceuticals Biosorption on Microbial and Algal-Derived Biosorbents

Zdravka Velkova ^{1,*}, Kristiana Lazarova ², Gergana Kirova ¹  and Velizar Gochev ^{2,*}

¹ Department of Chemical Sciences, Faculty of Pharmacy, Medical University—Plovdiv, 4002 Plovdiv, Bulgaria; gergana.kirova@mu-plovdiv.bg

² Department of Biochemistry and Microbiology, Faculty of Biology, Plovdiv University “P. Hilendarski”, 4000 Plovdiv, Bulgaria; klazarova@uni-plovdiv.bg

* Correspondence: zdravka.velkova@mu-plovdiv.bg (Z.V.); vgochev@uni-plovdiv.bg (V.G.)

Abstract: The removal of pharmaceuticals from aqueous environments has become a critical ecological challenge. Biosorption has emerged as a promising and cost-effective solution for pharmaceutical removal. This review examines the potential of microbial and algal-derived biosorbents, including fungi, bacteria, and algae, in the biosorption of pharmaceuticals from water. The removal efficiency of various types of biosorbents is discussed in relation to the chemical structure and functional groups presented on the biosorbent surfaces at various process parameters, such as pH, contact time, biosorbent dosage, and initial pharmaceutical concentration. Additionally, the benefits of chemical and physical modifications, immobilization techniques, and the reusability of biosorbents are highlighted. The major goal of the present review is not just to gather and discuss information about possible mechanisms of biosorption, which to some extent are still speculative, and to explain the effect of process parameters on the removal but also to highlight the advantages and disadvantages of various types of microbial/algal biosorbents and to ease the selection of proper biosorbents for pharmaceuticals removal. In this way, the review will benefit and induce more technological studies in the field of biosorption.

Keywords: biosorption; pharmaceuticals; removal



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1. Introduction

In recent years, Contaminants of Emerging Concern (CECs) have garnered significant attention due to their potential negative effects on both human health and ecosystems.

Sauvé and Desrosiers [1] determined that CECs are chemicals or materials “naturally occurring, manufactured or whose toxicity or persistence are likely to significantly alter the metabolism of a living being”. Many of these contaminants are not regulated under current environmental laws [2]. CECs can be classified into the following two main categories: chemical (chemicals) and biological (pathogens) contaminants [3]. Chemical contaminants include substances like pharmaceuticals and personal care products [4], pesticides [5], artificial sweeteners [6], micro- and nano-plastics, and the products resulting from their transformation [7]. Biological contaminants consist of antibiotic-resistant bacteria (ARB) [8] and antibiotic resistance genes (ARG) [9].

Over the past few years, special attention has been paid to pharmaceuticals due to their continuous use in human and veterinary medicine. Usually, the substances of concern are the pharmaceutical active compounds (PhACs), but their metabolites and degradation products may also be relevant, as well as some ingredients (excipients) other than the active ingredient, as well as the packaging material [10].

The occurrence of pharmaceuticals (steroid hormones) in river water was first reported in the early 1970s [11,12]. In a later stage, from the mid-1990s, the presence of pharmaceutical contaminants was seen as a new environmental problem, which can be explained not only by the increased production and use of pharmaceuticals but also by advancements in analytical techniques. New coupled methods, such as liquid chromatography (LC) and gas chromatography (GC) coupled with mass spectrometry (MS), have enabled the detection of pharmaceutical compounds at low concentrations in complex environmental matrices [13,14]. These developments have made it possible to identify pollutants that were previously difficult to detect. Environmental pollution with pharmaceuticals is related to the growth of their use, which is determined by the demographic aging of the population, the increasing number of chronic diseases, the availability and use of generics, and the development and use of new medicinal preparations [15,16]. In addition, large quantities of pharmaceuticals are used as veterinary medicine on farms worldwide, primarily to prevent and treat animal diseases, as well as to enhance economic benefits in intensive livestock production [17].

The release of pharmaceuticals into the environment occurs through various pathways (Figure 1) as follows: drug-containing waste from the pharmaceutical industry, hospitals, or households; wastewater and sewage sludge from municipal wastewater treatment plants; soil fertilization with organic fertilizers, including manure and slurry, or with other natural fertilizers containing animal excreta; and inadequate disposal of unused or expired medicines [18].

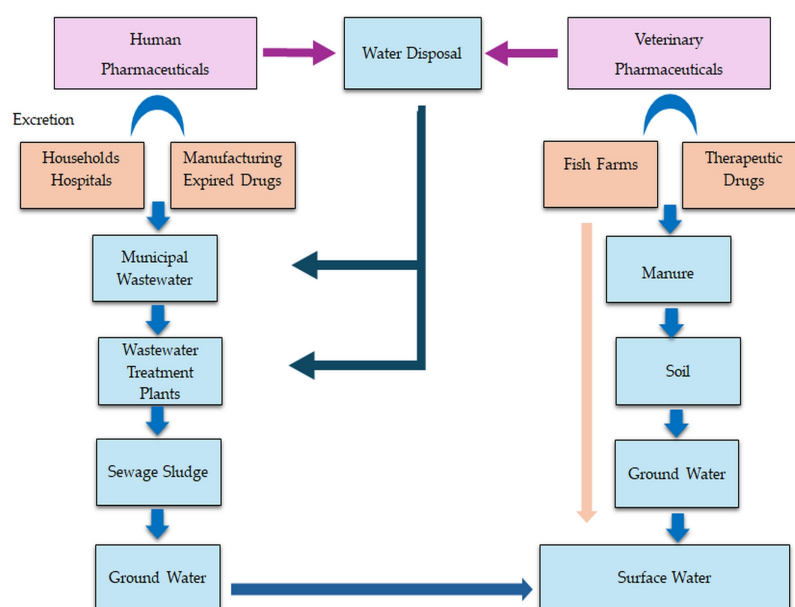


Figure 1. Various pathways of release of pharmaceuticals into the environment.

The concentrations of pharmaceuticals detected in various environmental matrices, such as wastewater, surface water, drinking water, and groundwater, vary widely—from very low (ng/L) to much higher levels (mg/L) [19–22]. The variability in concentration is influenced by factors such as the type of pharmaceutical, the local environment, and the specific pathways of contamination.

Pharmaceuticals are substances that have significant biological activity, even at very low concentrations. Therefore, when they are released into the environment, they risk interacting with organisms for which they were not intended. According to an OECD estimate, 10% of pharmaceutical products present a potential risk for the environment. In addition, drugs degrade very slowly and release active substances continuously until they

disappear. Once released into the environment, they persist for a long time in the form of traces, which could have medium- and long-term consequences that have not yet been studied [23].

The persistence of pharmaceuticals in the environment poses significant risks to both human health and aquatic life. The impact on human health is harder to quantify because exposure occurs indirectly, primarily through drinking water and the ingestion of pharmaceutical residues found in plant crops, fish, dairy products, and meat. This prolonged exposure can lead to potential health concerns, but the full extent of the risks remains challenging to assess due to the complexity of human exposure pathways [14].

Several studies have shown that some pharmaceuticals can have harmful effects on ecosystems. These impacts include mortality of aquatic organisms, as well as changes in physiology, behavior, and reproduction [23]. As an illustration, oral contraceptives, containing synthetic hormones such as estrogen and progestin, have caused the feminization of fish and amphibians, which affects the ability of their population to reproduce [24]. Antidepressants have modified the behavior of fish, making them more vulnerable to predators [25]. It has been shown that the presence of ibuprofen (0.32 mg/L) and carbamazepine (2.24 mg/L) in aquatic environments affects the morphology of *Hydra attenuata* (cnidarian) [26]. The decline in dung beetle populations has been attributed, at least in part, to the use of antiparasitic pharmaceuticals, including ivermectin, in livestock production [27].

It is also clear that the release of antibiotics into water and soil contributes to the acceleration of the development and spread of antibiotic-resistant bacteria and fungi. This resistance poses a significant threat to public health, as it can lead to the ineffectiveness of antibiotic treatments, making infections harder to treat and control [8,23,25].

Conventional wastewater treatment plants (WWTPs) were not designed to remove pharmaceuticals and, therefore, do not completely eliminate them from wastewater [28]. The effectiveness of removing pharmaceuticals varies widely and depends on the influent mass of the compounds, their physicochemical properties, the environment, and operating conditions [29]. For example, the removal efficiency of sulfamethoxazole in WWTPs can reach up to 50% depending on the treatment applied, while diclofenac is removed on average 48% [30,31]. According to Comber et al. [32], ethinyloestradiol, diclofenac, ibuprofen, propranolol, and macrolide antibiotics are present at high enough concentrations in the effluent of 890 wastewater treatment plants (13% of all) in the United Kingdom.

Different treatment methods have been studied and applied for the removal of pharmaceuticals—physical, chemical, biological, and hybrid [33]. Some of them are shown in Figure 2.

Along with their advantages, these techniques have several drawbacks, such as the generation of harmful products, high operational and maintenance costs, etc. [34–36].

Adsorption is a widely used process due to its simplicity of design, safety, and environmental nontoxicity; however, traditional adsorbents are often costly [37]. In recent years, the application of various low-cost adsorbents—inactive or dead biomasses (mainly from microbial, algal, and plant origin)—has received great attention. The major advantage of these biosorbents undoubtedly is the low cost. Additionally, the complex surface of microbial and algal biosorbents provides diverse chemical groups acting as binding sites, expanding the possibilities for the removal of a wide range of pharmaceutical wastes from aqueous solutions, even at concentrations when conventional methods are less effective.

This review focuses on the following concepts: (i) basic principles, methods, and techniques used in biosorption; (ii) pharmaceuticals as biosorbates; (iii) general characteristics of microbial and algal-derived biosorbents and their use in pharmaceutical biosorption; and (iv) future priorities. Attention is paid to process parameters, equilibrium, and kinetic studies.

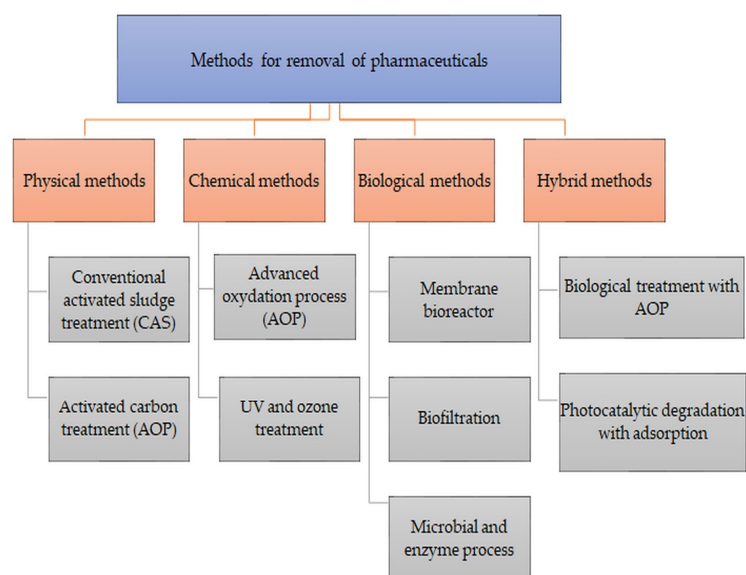


Figure 2. Methods for removing pharmaceutical contamination in the environment.

2. Biosorption—Basic Concepts

Biosorption is a simple, cost-effective, efficient, and eco-friendly method for treating wastewater. According to Gadd [38], biosorption may be simply defined as the removal of substances from solution by biological materials. These biological materials can be bacteria, cyanobacteria, fungi, algae, industrial and agricultural wastes, plant derivatives, and other polysaccharide materials. The biological materials (biosorbents) can be dead or living, while the sorbate includes metal ions, dyes, pharmaceuticals, etc.

Biosorption is classified into two types—active biosorption (bioaccumulation), which involves living cells, and passive biosorption (biosorption), when dead biomass is used [39]. The use of non-living biomass (such as microbial and algal biomass) offers several advantages over living biomaterials, including lower cost, no need for providing a nutrient medium or maintaining pure microbial cultures, no increase in the chemical oxygen demand (COD) of water, high biosorption and desorption rates, effective performance over a wide pH range, simplicity of equipment used, and rapid and easy regeneration of the biomass [40–42].

The cell wall of microbial and algal biomasses plays a crucial role in the removal of both inorganic and organic compounds. This is due to the presence of various functional groups with different charges and geometries, such as carboxyl, hydroxide, phosphate, amino, imidazole, sulfate, sulfhydryl, and others. As biosorption is a process that takes place on the adsorbate–adsorbent interface, the physical or chemical modification of the biomass has a significant effect on the biosorption capacity. Physical modification methods include dry heating at 45–80 °C, autoclaving at 121 °C for 15–20 min, lyophilization, etc. According to Salem [39], they are simple, fast, and inexpensive but usually do not significantly increase the biosorption capacity of biomass. Another way to enhance the biosorption capacity of biomass is through chemical modification. Different methods are used—treatment with solutions of acids, bases, ethanol, cationic or anionic surfactants, formaldehyde, introduction or “blocking of functional groups”, lipid extraction, etc. [40]. The efficiency of the physical and chemical modification methods depends on the origin of the biomass used, respectively, on its cell composition and functional groups present on the cell wall. The efficiency also depends on the physicochemical properties of pollutants to be removed and should be examined regarding each particular process.

Biomasses can be applied in the biosorption process of contaminants in free or immobilized forms. Immobilization provides good porosity, better mechanical stability, elasticity, higher resistance, improved separation, regeneration, and reusability. The immobilized biomass can be included in columns with fixed and fluidized beds [42–45]. In the immobilization of biomass, the following methods, like those for the immobilization of enzymes, are used: physical adsorption, covalent binding, cross-linking, incorporation within polymer matrices, and encapsulation. Velkova et al. [43] have outlined the benefits and drawbacks of widely used traditional immobilization methods.

The mechanism of biosorption depends on the type of biomass and the contaminant present in the sample and may be physical and chemical, but it is possible that several mechanisms/interactions can be present simultaneously [46,47]. The following four dominant mechanisms can be considered in the biosorption of pharmaceuticals on inactive microbial and algal biomasses: electrostatic attraction, π – π interaction, hydrogen bonding interaction, and physical adsorption in the pores of the biosorbent [46,48].

To obtain information on biosorbent characteristics and investigate the biosorption mechanism, various analytical techniques have been used (Figure 3) [49].

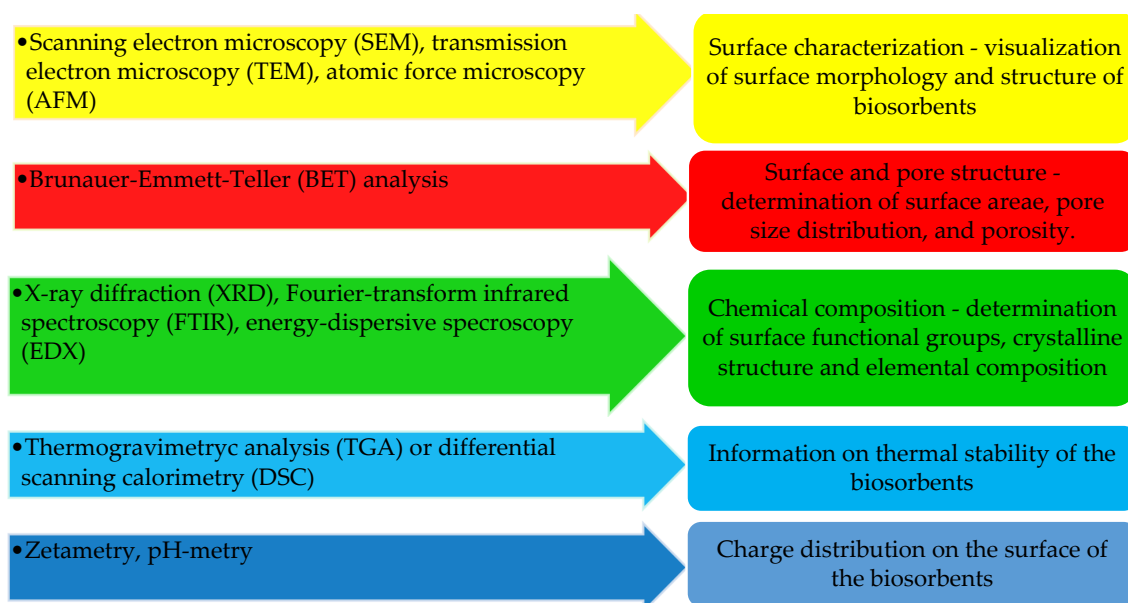


Figure 3. Analytical techniques used for biosorbent characterization and biosorption mechanism investigation.

Some of the techniques mentioned above can be costly for routine research, and the information they provide may not always be essential. However, different techniques offer distinct insights, so a more comprehensive interpretation of biosorption processes can be achieved by combining data from multiple methods [43,50].

The *biosorption* process can be performed under batch and dynamic (*continuous*) modes. The *batch* systems are the preferred option for laboratory-scale research; the main goal is to optimize the process parameters (solution pH, ionic strength, initial pollutant concentration, concentration of biosorbent, contact time, and temperature), determining the effectiveness of biosorption removal. The *dynamic* systems are more complex and are relevant for evaluating the application of the process to real samples. Numbers of mathematical models are available to evaluate the equilibrium and kinetics studies in batch mode [38,42,47] and the dynamics in biosorbent columns [44,45,47].

Reuse or regeneration of biosorbents is an important step in biosorption research. On the one hand, reuse determines the cost-effectiveness of the removal process, and on

the other hand, it can reveal the mechanism of biosorption. Unfortunately, relatively few studies have been conducted on this topic.

3. Pharmaceuticals Pollutants as Biosorbates

According to Bush [51], environmentally concerning groups of pharmaceuticals include anti-inflammatories and analgesics, antibiotics, antiepileptics, antidepressants, lipid-lowering agents, antihistamines, β -blockers, cytostatics, and other substances (barbiturates, narcotics, antiseptics, and contrast media).

Pharmaceuticals differ significantly in their chemical structure due to their structural complexity and the presence of different functional groups (carboxylic, amide, amino, hydroxyl, etc.). The chemical structure of pharmaceuticals influences their physicochemical properties (lipophilicity, aqueous solubility, degree of ionization, etc.) [13,46]. Many pharmaceuticals contain acidic and/or basic functional groups, allowing them to exist in anionic, cationic, neutral, or zwitterionic forms depending on the solution's pH. Therefore, optimizing this parameter is crucial and should be examined for each specific biosorbent–biosorbate system. Additionally, pharmaceuticals can vary in their hydrophobicity and reactivity, which influence the efficiency of the removal process.

4. Biosorbents Based on Microbial and Algal Biomasses for Pharmaceuticals Removal

Various biomasses, including bacteria, fungi, and algae, have been used as biosorbents for pharmaceutical removal from model water solutions and wastewater.

4.1. Bacteria-Derived Biosorbents

Bacteria have a complex cell wall with various biosorption centers (e.g., carboxyl, phosphonate, amine, and hydroxyl groups) and a high surface-to-volume ratio [52]. The cell wall, consisting of lipids, polysaccharides, and proteins, differs structurally between Gram-positive and Gram-negative bacteria. Gram-positive bacteria typically have a higher biosorption capacity due to their thicker peptidoglycan layer [53,54]. Various bacterial biomasses have been successfully used to remove toxic metal ions and dyes from model solutions and wastewater [41,42,46,52,55].

Al-Gheethi et al. investigated cephalexin (CFX) removal from aqueous solution by a consortium of bacterial living and dead cells (*Burkholderia cepacia*, *Chrysomonas luteola*, *Pseudomonas fluorescens*, *Bacillus subtilis*, *Bacillus megaterium*, *Bacillus stercorophilus*, *Citrobacter freundii*, *Kluyvera* spp.), which are tolerant to antibiotics [56]. The factors affecting the biosorption process, such as the initial concentration, biomass dosage, pH, contact time, and temperature, were studied. The highest removal efficiency at a high CFX concentration (5 mg/mL) was achieved by dead cells (82.36%) compared to living cells (46.66%). However, the reverse effect was observed at a 0.4 mg/mL initial concentration, with dead cells achieving 92.98% removal, while living cells reached 94.73%. Living cells demonstrated greater efficiency than dead cells at a pH range of 4 to 6. The study also found that the removal efficiency of CFX decreased in the presence of Ni^{2+} , Cu^{2+} , and Pb^{2+} ions.

Ethanol-pretreated waste *Streptomyces fradiae* biomass was used as a biosorbent to remove tetracycline (TET) from model aqueous solutions [57]. FTIR spectra of the pretreated biomass, both before and after biosorption, revealed that the primary functional groups involved in tetracycline removal were hydroxyl, carboxyl, and/or amine groups. The uptake of the pollutant remained relatively constant within a pH range of 4.0 to 6.0, with removal efficiencies varying between 58.12% and 57.5%. The biosorption likely occurred due to electrostatic attraction between the positively charged biomass and the zwitterionic tetracycline species.

A magnetic biosorbent was created by immobilizing *Escherichia coli* onto nano-sized magnetic iron oxide, and its effectiveness in diclofenac (DCF) biosorption was evaluated. The magnetic biocomposite was characterized using various analytical techniques, including FTIR, SEM, EDX, and vibrational sample magnetometry (VSM). The FTIR spectrum of the magnetic biosorbent showed the presence of several functional groups, including carboxyl, phosphate, and amino groups. The results of SEM and EDX analysis indicated the successful immobilization of *Escherichia coli* onto Fe₃O₄ nanoparticles. The magnetic saturation value, 63 emu/g, revealed that the immobilized biomass can be isolated from treated samples using a sample magnetic field. The effects of contact time (5–60 min), temperature (30 °C, 40 °C, and 50 °C), biosorbent dose (1–5 g/L), and initial DCF concentration (50–200 mg/L) on the biosorption process were investigated. It was observed that the removal of DCF occurred quickly—it reached 86.9% after 5 min of contact time at pH 6.2, 100 mg/L initial biosorbate concentration, and biosorbent dosage of 5 g/L. After 30 min of contact time, the calculated removal efficiency was 93.96%. The maximum biosorption capacity, according to the Langmuir model, was 46.01 ± 0.12 mg/g. Thermodynamic data indicated that the biosorption of DCF onto the biocomposite was endothermic. The proposed mechanism for DCF biosorption involved hydrogen bonding interactions, π – π interactions, and electrostatic attraction [58].

Gopal et al. proposed a hybrid method for removing TET by combining biosorption on a bionanocomposite (BNC) with a photo-assisted process [59]. The BNC was created by immobilizing Fe₃O₄ nanoparticles, TiO₂ (P25) nanoparticles, and dead biomass of *Acinetobacter* sp. (a tetracycline-resistant bacterium) in alginate beads. The biosorbent was characterized using SEM/EDX, TEM, and FTIR techniques. Control studies with different configurations showed varying TET removal efficiencies, with the highest removal ($98 \pm 0.5\%$) achieved when biosorption was coupled with UV-C irradiation. The process was tested with natural water systems (tap, groundwater, and lake water) spiked with 10 mg/L tetracycline. The ecotoxicity of treated effluents was assessed using cyanobacterial strains (*Chlorella* sp. and *Scenedesmus* sp.) and bacterial strains (*Pseudomonas aeruginosa* and *Escherichia coli*), showing minimal toxic effects. Additionally, the reusability of the bionanocomposite was demonstrated up to the 4th cycle. The study concluded that the nanocomposite-loaded alginate bead-based photocatalytic removal of TET resulted in significantly fewer toxic by-products, enhanced reusability, and the ability to be easily separated after the process.

The removal of ethacridine lactate (EL) was investigated with a biosorbent prepared by immobilizing a thermally deactivated *Lactococcus lactis* biomass onto calcium alginate beads [60]. The efficiency of biosorption was evaluated in batch mode. Equilibrium was achieved within 24 h, with a removal efficiency of 90.72% observed when the initial concentration of EL was 60 mg/L. The regeneration of the biosorbent was possible with 0.05 M HCl. Different analytical techniques (SEM and FTIR) suggested the retention of EL by the biosorbent and established a possible interaction between them.

Cyanobacteria (formally known as blue-green algae) are also applied for biosorption and removal of pharmaceuticals from contaminated matrices. It should be noted that many cyanobacteria can be cultivated on a large scale, so a low-cost biomass can be produced and used in the biosorption process [61].

Usually, biosorbents can be chemically modified in order to improve their biosorption capacity. Momin et al. [62] studied the biosorption of DCF by dead *Microcystis aeruginosa* biomass pretreated with KOH. The maximum biosorption capacity, determined using the Langmuir model, was 11.55 ± 0.64 mg/g. The modified biomass demonstrated similar efficiency in removing DCF from binary solutions containing Cd²⁺ ions (1–100 mg/L) and ibuprofen (5 mg/L). FTIR results showed that the functional groups –OH, –COOH, and

–C=C– on the biomass surface played a key role in DCF biosorption. Thermodynamic studies confirmed that the biosorption process was both exothermic and spontaneous. The biosorbed DFC could be effectively desorbed using 99.8% methanol over three biosorption–desorption cycles.

Live and dead biomass of the unicellular *Phaeodactylum tricornutum* were tested for biosorption of ibuprofen (IBU). The removal efficiency for both biomasses was similar (99.9%) for the initial IBU concentration from 0.1 to 2.5 mg/L, at pH 8.2 and at 0.8 g/L biomass dosage [63].

Cyanobacteria belonging to the genus *Scenedesmus* have been the subject of research by several authors. Alkaline-modified biomass *Scenedesmus obliquus* was used for the biosorption of tramadol (TRAM) and other pharmaceuticals—cefadroxil (CEFA), paracetamol (PAR), ciprofloxacin (CIP), and IBU [64]. A batch biosorption study was conducted to evaluate the impact of various operational parameters, including pH, initial concentration, and contact time, on the removal process of TRAM. The removal efficiency for 50 mg/L TRAM was 91% at pH 7, with a contact time of 45 min and a biosorbent dose of 0.5 g/L. It was reported that hydrophobic interactions between the amino and carbonyl groups in TRAM and the hydroxyl and carbonyl functional groups on the surface of the biosorbent may be responsible for the biosorption process. At pH 7.8 (the natural pH of the solution), the removal of TRAM in the presence of CFX, PAR, IBU, and CIP was investigated. At this pH, CEFA ($pK_a = 3.45, 7.03, 9.6$) exhibited the highest removal rates, followed by PAR ($pK_a = 9.5$), due to their higher pK_a values, compared to IBU ($pK_a = 4.91$) and CIP ($pK_a = 6.2$). Additionally, the modified algal biomass demonstrated high reusability potential, with a 4.5% decrease in removal efficiency after 3 runs.

Dead biomasses of *Scenedesmus* sp. (Chlorophyceae) and *Synechocystis* sp. (Cyanophyceae) were evaluated for the biosorption of DCF; the calculated maximum biosorption capacities, from the Langmuir isotherm model, were 28 and 20 mg/g, respectively [65].

Silva and co-workers [66] studied the biosorption of two non-steroidal anti-inflammatory drugs (salicylic acid and IBU) on *Scenedesmus obliquus* biomass. The equilibrium data were best described by the Langmuir isotherm model, with maximum adsorption capacities of 60 mg/g and 12 mg/g for salicylic acid and IBU, respectively. The biosorption of the two target drugs was favorable, spontaneous, and exothermic.

The biosorption of a mixture of four widely used veterinary pharmaceutical products, TET, CIP, sulfadiazine (SDZ), and sulfamethoxazole (SMX), was evaluated using a *Scenedesmus almeriensis* microalgae-bacteria consortium, recovered from a high-rate algae pond (HRAP) [67]. Biomass was inactivated through lyophilization. SEM and FTIR were used to examine the surface morphology and to identify the effect of the antibiotic functional groups on the consortium's surface. TET and CIP showed a stronger affinity to the consortium. At an initial concentration of 20 µg/L, the removal efficiencies were 80% for TET, 30% for SDZ, and 30% for SMX; CIP was completely removed after 24 h. The maximum biosorption capacities, determined using the Langmuir isotherm model, were 0.83, 0.52, 0.26, and 0.01 µg/mg for TET, CIP, SDZ, and SMX, respectively.

Live and dead biomass *Chlorella vulgaris* was used to remove an anticancer drug, named flutamide (FLU), from wastewater [68]. The maximum biosorption capacity of the living biomass (26.8 mg/g) was higher than that of the dead biomass (12.5 mg/g). Maximum FLU removal was observed at low pH values. The low biosorption removal at neutral and basic pH values was attributed to an electrostatic repulsion between the negatively charged biomass (deprotonation of carboxylic, phosphoric, and amine groups) and the negatively charged FLU species.

Mirizadeh et al. [69] reported the preparation of two biocomposites composed of magnetite, chitosan, and *Chlorella vulgaris* (MCC) or *Arthrospira platensis* (MCA). The magnetized

composites (MCC and MCA) displayed more porous and rougher surfaces compared to the composites without magnetite. In addition, the MCA surface exhibited more cracks and fine pores that could potentially improve its antibiotic biosorption capacity compared to MCC. The magnetic biocomposites were evaluated for the biosorption of TET, CIP, and amoxicillin (AMX), showing a descending trend in the maximum biosorption capacity in the following order: TET > CIP > AMX. The biosorption process was effective for up to four cycles, with a minimal loss in biosorption capacity at the following equilibrium: no more than 7% for TET, 4% for CIP, and 9% for AMX.

The biosorption of metronidazole (MNZ) using *Spirulina platensis* microalgae biomass from aqueous solutions was investigated by Esmali et al. [70]. They used Response Surface Methodology (RSM) and the Box–Behnken design (BBD) to optimize operational parameters, including initial MNZ concentration (10–80 mg/L), pH (4–10), contact time (10–60 min), and biomass dose (0.1–0.5 g/L). The biomass achieved 88.15% removal of MNZ under the following optimized conditions: a contact time of 38.05 min, a pH of 7.71, and a biomass dose of 0.3 g/L.

Escapa et al. [71] examined the potential of a dead biomass of the cyanobacteria *Synechocystis* sp. in the biosorption of PAR from contaminated water. Results were compared with the biosorption of PAR onto a commercial activated carbon under the same experimental conditions. The calculated maximum biosorption capacity of the biomass was around five times smaller than that of the activated carbon.

Dead biomass *Chlorella* sp. (NLB) and its by-product after lipid extraction (LEB) were compared for the removal of TET and minocycline (MC) from aqueous solutions [72,73]. The results showed that LEB removed TET more efficiently than the dead biomass, achieving 76.9% removal at an initial concentration of 40 mg/L and a biomass dosage of 1.6 g/L. A multilayer mechanism on a heterogeneous surface was suggested for TET removal. For MC, NLB removed $90.8 \pm 1.3\%$ at an initial concentration of 53.89 mg/L with 50 mg of biomass at pH 10, while LEB removed $80.8 \pm 1.4\%$ under the same conditions.

Table 1 shows maximum biosorption capacities, calculated from the Langmuir model in mg/g, and operating conditions for different bacterial biomass-derived biosorbents.

Table 1. Bacterial biomass-derived biosorbents for pharmaceutical removal.

Biosorbent	Biosorbate	Experimental Conditions	q _{max}	Ref.
<i>Arthrospira platensis</i> / magnetite / chitosan	Tetracycline	25 °C, pH = 8.0, C ₀ = 10–200 mg/L, W = 0.05 g/L, 2 h	834.0	[69]
	Ciprofloxacin	25 °C, pH = 6.0, C ₀ = 10–200 mg/L, W = 0.1 g/L, 2 h	394.9	
	Amoxicillin	25 °C, pH = 6.0, C ₀ = 10–200 mg/L, W = 0.1 g/L, 2 h	150.8	
<i>Chlorella vulgaris</i> / magnetite / chitosan	Tetracycline	25 °C, pH = 8.0, C ₀ = 10–200 mg/L, W = 0.05 g/L, 2 h	831.1	[69]
	Ciprofloxacin	25 °C, pH = 6.0, C ₀ = 10–200 mg/L, W = 0.1 g/L, 2 h	374.2	
	Amoxicillin	25 °C, pH = 6.0, C ₀ = 10–200 mg/L, W = 0.1 g/L, 2 h	140.2	
<i>Escherichia coli</i> / magnetite / chitosan	Diclofenac	30 °C, pH = 6.2, C ₀ = 50–200 mg/L, W = 5 g/L, 0.5 h	46.01	[58]
<i>Microcystis aeruginosa</i> (chemically modified)	Diclofenac	28 °C, pH = 6.8, C ₀ = 1–10 mg/L, W = 1 g/L, 2 h	11.55	[62]
<i>Phaeodactylum tricornutum</i>	Ibuprofen	38 °C, pH = 8.2, C ₀ = 0.1–15 mg/L, W = 0.8 g/L, 6 h	3.96	[63]
<i>Scenedesmus obliquus</i> (chemically modified)	Tramadol	pH = 7.0, C ₀ = 25–200 mg/L, W = 0.5 g/L, 0.75 h	140.25	[64]
<i>Spirulina platensis</i>	Metronidazole	pH = 7.7, C ₀ = 10–150 mg/L, W = 1.0 g/L, 0.3 h	20	[70]
<i>Synechocystis</i> sp.	Paracetamol	25 °C, pH = 7.0, C ₀ = 15–150 mg/L, W = 1 g/L, 2 h	53	[71]
<i>Streptomyces fradiae</i> (chemically modified)	Tetracycline	25 °C, pH = 5.0, C ₀ = 25–100 mg/L, W = 2 g/L, 2 h	38.61	[57]

W—biosorbent concentration, g/L; C₀—initial concentration, mg/L; contact time, h; temperature, °C.

4.2. Fungal-Derived Biosorbents

Fungi are a large and diverse group of eukaryotic microorganisms. The cell wall is composed mainly of glucans, chitin, chitosan, and glycoproteins and can make up 30% or more of the dry weight of the fungus [74]. Yeasts, molds, and mushrooms have major practical importance in the field of biosorption [75,76].

4.2.1. Yeasts

The yeast cell wall consists of various organic compounds and their polymers, including glucan (28%), mannan (31%), proteins (13%), lipids (8%), and chitin and chitosan (2%). The composition and organization of the cell wall can vary across different yeast genera, including *Saccharomyces*, *Candida*, *Kluyveromyces*, *Yarrowia*, and *Schizosaccharomyces* [77]. Yeast biomasses, in particular, belonging to the genus *Saccharomyces*, have received increasing attention in biosorption studies. Yeasts are unicellular, fast-growing, and easily cultivated on a large scale. In addition, yeast biomasses can be sourced as a by-product from various food and beverage industries. Yeast-derived biosorbents have received considerable attention in the biosorption of metal ions and dyes [43,74]. Free and immobilized yeast biomasses have been investigated as biosorbents of pharmaceuticals.

Saccharomyces cerevisiae biomass, inactivated at 60 °C for 12 h, was tested for AMX removal [78]. The maximum removal efficiency reached 93% under the following optimal conditions: pH 5, initial AMX concentration of 5 mg/L, biosorbent concentration of 0.75 g/L, contact time of 90 min, and temperature of 25 °C. The biosorption process occurred through attractive forces between the AMX and the biosorbent. The experimental data best fitted the Freundlich isotherm model, and the pseudo-second-order kinetic model accurately described the biosorption kinetics.

Santos and colleagues explored the use of chemically activated *Saccharomyces cerevisiae* (baker's yeast) biomass for the biosorption of IBU from aqueous solutions [79]. The yeast biomass was activated by extraction with hexane and methanol using a Soxhlet extractor. The biosorption of IBU was highly dependent on pH and temperature, with optimal removal rates observed at pH 2.0 and a temperature of 40 °C. FTIR analysis revealed that the biosorption of IBU was mediated by functional groups on the yeast, including carboxyl, hydroxyl, phosphoryl, and amino groups.

Yeast biomass (YB) residue from ethanol production was inactivated by a spray-drying technique and evaluated as a biosorbent for 17 α -ethinylestradiol (EE) alone and in the presence of estrone (EST) [80]. Using a chemometric approach, the best conditions for EE removal were identified as pH 10, a biosorbent mass of 0.5 mg/L, and an ionic strength of 0.1. In binary solutions (EE and EST), the biosorption behavior was associative and competitive, depending on the concentrations of EE and EST. This competitive behavior is attributed to the similar chemical structures of EE and EST and their comparable organic carbon–water partition coefficients (K_{oc}).

Rusu et al. investigated the immobilization of three yeast biomasses (*Saccharomyces cerevisiae*, *Saccharomyces pastorianus*, and residual *Saccharomyces* biomass) in calcium alginate and chitosan for biosorption of rifampicin (RIF), CFX, and EL from model solutions [81–84]. The highest biosorption capacities for RIF were achieved using *Saccharomyces cerevisiae*—chitosan (SC-C-2.5%) and *Saccharomyces pastorianus*—chitosan (SP-C-2.5%) biocomposites, with uptake values of 24.70 and 24.84 mg/g, respectively, at pH 6, 50 mg/L RIF, 1 g/L biosorbent, and 12 h of contact time. Electrostatic interactions, π – π interactions, and hydrogen bonding were suggested as the main mechanisms of biosorption [81]. The biosorption of CFX using *Saccharomyces cerevisiae*—calcium alginate composite beads was strongly pH-dependent, with maximum removal efficiency (58.56%) observed at pH 4 [82]. The two types of biosorbents based on *Saccharomyces pastorianus* (dried and waste biomass)

immobilized in calcium alginate showed superior removal efficiencies (>90%) for all EL concentrations (20–60 mg/L) at an initial pH of 4.0 and biosorbent dose of 2 g/L [83]. In fixed-bed column studies, biosorption capacity was influenced by functional parameters such as bed height, EL concentration, and flow rate [84].

Chitosan was used as a support for encapsulating *Candida* sp. biomasses (*Candida lipolytica*, *Candida membranaefaciens*, *Candida tropicalis*, and *Candida utilis*) to get biocomposites for the removal of TET. The maximum biosorption capacity for the biocomposite *Candida utilis*—chitosan was 2.309 mg/g (pH 4, TET concentration from 10 to 100 mg/L, at 25 °C) [85].

4.2.2. Molds

Molds are filamentous fungi; their mycelium, a tangled network, is composed of fine, branching filaments called hyphae [86]. Of interest in the biosorption of pharmaceuticals are micromycete biomasses, mainly from the genus *Rhizopus*.

Aksu and Tunç [87] studied the biosorption of penicillin G using the following three biosorbents: dried *Rhizopus arrhizus* biomass, dried activated sludge, and commercial activated carbon. The study was conducted in batch mode to evaluate the efficiency of biosorption. The optimal pH and temperature for biosorption were found to be 6.0 and 35 °C for all three biosorbents. Under the optimal conditions, the maximum uptake capacities were calculated as follows: 591.8 mg/g for dried *Rhizopus arrhizus* biomass, 434.8 mg/g for dried activated sludge, and 463.0 mg/g for commercial activated carbon.

The biosorption of naproxen, a non-steroidal anti-inflammatory drug, using *Rhizopus oryzae* biomass (wet and dried) was investigated by Melgoza and co-workers [88]. The study highlights that pH plays a critical role in the biosorption of naproxen, with alkaline conditions favoring wet biomass and mildly acidic conditions (pH 4.7) being optimal for dried biomass. Electrostatic interactions were identified as the key mechanism driving the biosorption process.

The study by Azamateslamtalab et al. [89] focused on the removal of TET from aqueous solutions using raw and pretreated *Rhizopus oryzae* biomass. The biomass was modified using a 0.2 M NaOH solution in an ultrasonic device (30 min, 40 kHz), followed by washing, autoclaving, and drying. Using various analytical techniques, it was found that this surface modification leads to an increase in the specific active surface area, the number of active functional groups associated with TET removal, the roughness, and furrows in the surface of the fungal biomass. This agreed with the results, which showed an improvement in the maximum biosorption capacity using the pretreated biomass in comparison with the raw biomass (67.93 vs. 38.02 mg/g). The removal was studied in the presence of Na⁺ and Ca²⁺ ions, and it was observed that with an increase in their concentration, the removal efficiency of TET decreased. Based on FTIR spectra, before and after biosorption tests, π – π or cation– π interaction mechanisms for tetracycline adsorption can be suggested.

Recently, Azamateslamtalab et al. [90] reported the modification of two fungal biomasses, *Rhizopus oryzae* and *Penicillium citrinum*, with Fe₃O₄ magnetic nanoparticles for the biosorption of TET from aqueous solutions. FTIR and XDR analysis confirmed the successful incorporation of the nanoparticles into the fungal biomasses. Superparamagnetic behavior was observed for *Rhizopus oryzae*/Fe₃O₄ and for *Penicillium citrinum*/Fe₃O₄ with a saturation magnetization of 38.63 and 35.47 emu/g, respectively, which means they can be easily manipulated using an external magnetic field, allowing for their efficient separation from treated water. The modified fungal biomasses showed a mesoporous structure, and the BET surface area exceeded 40 m²/g, further enhancing their biosorption capacity. The study found that as contact time and biosorbent dosage increased, the removal efficiency also increased. However, higher TET concentrations and temperature led to a decrease

in efficiency. The maximum adsorption capacities for TET were 31.53 mg/g for *Rhizopus oryzae*, 42.98 mg/g for *Penicillium citrinum*, 65.44 mg/g for *Rhizopus oryzae*/Fe₃O₄, and 71.49 mg/g for *Penicillium citrinum*/Fe₃O₄. The biosorption process was characterized as spontaneous and exothermic.

More recently, *Aspergillus flavus* biomass was modified using magnetic nanoparticles (MnFe₂O₄) and the metal–organic framework ZIF-67 for TET biosorption [91]. The modifications were confirmed by physicochemical tests. It was found that *Aspergillus flavus*-MnFe₂O₄-ZIF-67 has a significantly higher surface area (139.83 m²/g) compared to *Aspergillus flavus*/MnFe₂O₄ (17.504 m²/g) and *Aspergillus flavus* (0.786 m²/g). The composite biosorbent *Aspergillus flavus*-MnFe₂O₄-ZIF-67 exhibited the highest removal efficiency of 99.04% at pH 7, with an initial TET concentration of 10 mg/L, a biocomposite mass of 1 g/L, and a contact time of 40 min. Under optimal conditions, the maximum biosorption capacity was calculated to be 43.87 mg/g. Both composite biosorbents, *Aspergillus flavus*-MnFe₂O₄-ZIF-67 and *Aspergillus flavus*-MnFe₂O₄, could be separated from the solution using an external magnetic field. The reusability of these biosorbents was tested over 8 cycles.

Jureczko and Przysaś [92] investigated the biosorption potential of biomasses from white-rot fungi for the removal of two cytostatic drugs, bleomycin and vincristine. The selected drugs are excreted unmetabolized in high rates and poorly degraded in WWTP [93]. The study included *Fomes fomentarius* (CB13), *Hypholoma fasciculare* (CB15), *Phyllotopsis nidulans* (CB14), *Pleurotus ostreatus* (BWPH), and *Trametes versicolor* (CB8), representing different white-rot fungi species. The research was carried out on different types of biomasses—alive and dead (autoclaved). In the biosorption studies conducted with live biomass, the highest removal efficiency of bleomycin was 23% using *T. versicolor* (CB8), and for vincristine—20% using *P. nidulans* (CB14). Dead *T. versicolor* (CB8) showed higher biosorption abilities for both drugs, with removal rates of 35% for bleomycin and 17% for vincristine [92].

The ability of live and dead biomass of the white-rot fungus *Trametes hirsuta* to remove a mix of 17 pharmaceuticals at low concentrations (20–500 ng/L) was examined [94]. The biomass was inactivated using the following two methods: a heat-kill treatment (autoclaving for 45 min at 121 °C and 19 psi) and a biocide treatment (10 mM NaN₃ solution). The removal of pharmaceuticals involved the following three mechanisms: biosorption, extracellular enzyme-mediated biodegradation, and intracellular enzyme-facilitated biodegradation. The same pharmaceuticals, including ketoprofen, acetaminophen, AMX, and iopromide, were more effectively removed in the presence of live biomass, indicating the role of extracellular enzymes. The removal rates of mefenamic acid, IBU, indomethacin, and carbamazepine were similar with both live and autoclaved biomass. Only fenofibrate was completely removed by both living and dead biomass, whether autoclaved or NaN₃ pretreated. The higher removal rates with autoclaved biomass were attributed to changes in the cell membrane or its breakdown, which may facilitate the removal process.

Lucas et al. [95] evaluated the contribution of sorption processes to the elimination of four pharmaceuticals (carbamazepine, DCF, iopromide, and venlafaxine) during fungal treatment of wastewater. The study involved six different dead fungal biomasses, including three white-rot fungi (*Trametes versicolor*, *Irpex lacteus*, and *Ganoderma lucidum*) and three decomposing fungi (*Stropharia rugosoannulata*, *Gymnopilus luteofolius*, and *Agrocybe erebia*), tested in batch mode. The contribution of sorption processes to the overall removal of pharmaceuticals ranged from 3% to 13%, with different fungi showing varying degrees of sorption. For example, *S. rugosoannulata* contributed 4%, while *G. lucidum* showed a higher sorption contribution of 26%. The contribution of biosorption to the removal of pharmaceuticals is dependent on the type of fungal species. According to Nguyen et al. [96], hydrophilic pharmaceuticals showed negligible removal rates by both active and

inactivated fungi, which aligns with the lower sorption observed in this study for such compounds.

4.2.3. Mushrooms

Mushrooms are macro-fungi with tough texture and other physical characteristics that favor their application as biosorbents, so residues from mushroom production can be used as effective and inexpensive biosorbents.

Shiitake (*Lentinula edodes*) and champignon (*Agaricus bisporus*) stalks, as well as shiitake substrate (the medium in which shiitake was cultivated), were dried, ground, characterized, and tested for their ability to remove EE and PAR from contaminated water [97]. The porosity and pH value of the biosorbents were found to significantly influence the success of biosorption. Shiitake stalks exhibited the highest percentage of porosity, closed pores, and degree of deacetylation of chitosan, which contributed to their biosorption characteristics. Despite the favorable properties of shiitake stalks, champignon stalks were more effective in removing the selected pharmaceuticals from contaminated water.

Sarikaya et al. [98] investigated the biosorption of three tetracycline antibiotics — chlortetracycline (CTC), doxycycline (DC), and TET—from aqueous solutions using a biosorbent prepared from *Lactarius deliciosus*, an edible wild mushroom species. The impact of various factors was assessed, including biosorbent amount (0.01–0.1 g), pH (3.0–8.0), initial antibiotic concentration (30–300 mg/L for CTC and DC and 5–50 mg/L for TET), contact time (2–120 min), and temperature (7 °C, 16 °C, and 25 °C). It was found that the biosorption uptake of all tetracyclines decreased with increasing temperature, indicating that the biosorption process is exothermic. The biosorption process was effective in removing tetracyclines from tap and drinking water samples spiked with them. According to the FTIR analysis, the presence of carboxyl, hydroxyl, and amine groups on the *Lactarius deliciosus* cell wall was identified as playing a key role in the biosorption process.

Table 2 shows the maximum biosorption capacities (in mg/g, calculated from the Langmuir model) of different fungal-derived biosorbents and operational conditions.

Table 2. Fungal biomass-derived biosorbents for pharmaceuticals removal.

Biosorbent	Biosorbate	Experimental Conditions	q _{max}	Ref.
<i>Saccharomyces cerevisiae</i> / <i>calcium alginate</i>	Cephalexin	pH = 4.0, C ₀ = 10–80 mg/L, W = 1 g/L, 12 h	94.33	[82]
<i>Saccharomyces cerevisiae</i> (chemically activated)	Ibuprofen	40 °C, pH = 2.0, C ₀ = 5–35 mg/L, W = 1 g/L, 1 h	13.39	[79]
<i>Penicillium citrinum</i> PTCC 5304/Fe ₃ O ₄	Tetracycline	pH = 6.0, C ₀ = 10–100 mg/L, W = 1 g/L, 1 h	71.49	[90]
<i>Rhizopus arrhizus</i>	Penicillin G	25 °C, pH = 6.0, C ₀ = 50–1000 mg/L, W = 1 g/L, 24 h 35 °C, pH = 6.0, C ₀ = 50–1000 mg/L, W = 1 g/L, 24 h	558.7 591.8	[87]
<i>Rhizopus oryzae</i> PTCC 5263/Fe ₃ O ₄	Tetracycline	pH = 5.0, C ₀ = 10–100 mg/L, W = 1 g/L, 1 h	65.44	[90]
<i>Lactarius deliciosus</i>	Chlortetracycline	25 °C, pH = 4.0, C ₀ = 30–300 mg/L, W = 0.2 g/L, 2 h	216.4	[98]
	Doxycycline	25 °C, pH = 3.0, C ₀ = 30–300 mg/L, W = 0.2 g/L, 2 h	121.2	
	Tetracycline	25 °C, pH = 7.0, C ₀ = 5–50 mg/L, W = 0.2 g/L, 2 h	23.2	

4.3. Algae

Algae, especially marine algae, are another type of biomass applied for biosorption and removal of pollutants from contaminated effluents due to their availability in large quantities and at low cost.

Marine algae are classified into the following three main subgroups: brown algae (*Phaeophyta*), red algae (*Rhodophyta*), and green algae (*Chlorophyta*). The cell walls of these algae are composed of different substances that contribute to their biosorption properties. The cell walls of the brown algae typically contain cellulose, alginic acid, and guluronic acid, which are rich in carboxyl and sulfated groups. Brown algae are the most studied in the field of biosorption. Red algae also contain cellulose, along with sulfated polysaccharides such as agar and carrageenan, which contribute to their biosorption potential. The cell walls of green algae are primarily made up of cellulose, with a high percentage of proteins bound to polysaccharides, forming glycoproteins [99,100].

Navarro and co-workers [101] investigated the biosorption of two sulfa drugs, SMX and sulfacetamide (SAM), using marine algae (*Lessonia nigrescens* Bory (L13) and *Macrocystis integrifolia* Bory (S12)) from model water solutions. The biosorption process was found to be pH dependent, with maximum biosorption capacity occurring at pH values of 6 for SAM and 7 for SMX. Hydrogen bonding between the sulfa drugs and the algal biomass was suggested as the main mechanism responsible for the biosorption of the contaminants.

Recently, Khamayseh and Kidak [102,103] investigated the use of *Pithophora* macroalgal biomass for the biosorption of levofloxacin (LVX) and AMX from aqueous solutions. The maximum removal of LVX was achieved under the following optimized conditions: a temperature of 25 °C, an initial antibiotic concentration of 150 mg/L, a biosorbent dose of 0.5 g, and pH 6.5 [102]. For AMX, under optimal conditions (pH 5, biomass dose of 5 g/L, and a temperature of 25 °C), the biosorption capacity increased significantly from 1.52 mg/g at an initial AMX concentration of 10 mg/L to 18 mg/g at an initial AMX concentration of 150 mg/L [103].

Chemically modified *Posidonia oceanica* biomass (Neptune grass) was studied for its ability to remove oxytetracycline (OTC) from aqueous solutions [104]. The biomass was thoroughly analyzed using various techniques, including Boehm acid-base titration, pH at the point of zero charge (pH_{PZC}), FTIR, Solid-state Cross-Polarization Magic Angle Spinning Carbon-13 Nuclear Magnetic Resonance (¹³C CP-MAS NMR), optical microscopy, and TGA. The bi-Langmuir model, which assumes biosorption occurs at two localized sites, was applied to the experimental data. The maximum biosorption capacities for the two sites were found to be 11.8 mg/g and 4.4 mg/g under the following conditions: pH 6.0, a contact time of 1 h, initial OTC concentrations ranging from 9.2 to 80.6 mg/L, and a biomass concentration of 8 g/L. The biosorption process was determined to be spontaneous, exothermic, and entropically favorable. The formation of hydrogen bonds between the surface hydroxyl and phenolic groups of the biosorbent and the OTC molecule was identified as a key mechanism responsible for the biosorption.

Residual algal biomass, which remains after the extraction of value-added products (such as bioactive compounds or lipids), can be effectively used as a biosorbent.

Coehlo and co-workers [105] investigated the capacity of residual biomass from the brown algae *Sargassum filipendula*, after alginate extraction, to remove propranolol hydrochloride (PRO) from water solutions. The analysis of infrared spectra before and after biosorption revealed that the key binding sites involved in the biosorption process were carboxylic and hydroxyl groups on the algal biomass. The biosorption process was found to be spontaneous and exothermic.

López-Miranda et al. [106] utilized a *Sargassum* spp.-based system for the removal of commonly used drugs during the COVID-19 pandemic, including DCF, IBU, and PAR. The biomass, consisting of 59.3% *S. fluitans*, 35.8% *S. natans* I, 3.3% benthic microphytes, and 1.4% *S. natans* VII, was thermally and chemically pretreated and used as a natural precursor for the synthesis of carbon dots. The removal efficiencies for the drugs were 98% for DCF, 84% for IBU, and 54% for PAR. The system maintained its efficiency across

different pH values and could be reused for up to three cycles without significant loss in removal capacity. Structural and morphological modifications in the algae were observed using X-ray diffraction, FTIR spectroscopy, TGA analysis, and SEM, and the presence of the biosorbed drugs was confirmed.

The biochar derived from the pyrolysis of algal biomass has attracted attention in the biosorption of pharmaceuticals. The interest is due to its porous structure, large surface area, and abundant active sites [107]. In addition, it can be easily modified by physical and chemical methods to optimize its biosorption capacity [108].

Pimentel-Almeida et al. [109] studied the biosorption of acetaminophen using biochar derived from *Sargassum cymosum* macroalgae. The waste beach-cast *Sargassum cymosum* biomass was converted into biochar through pyrolysis at 800 °C in a limited oxygen atmosphere. The biochar was characterized and tested for acetaminophen removal in both batch and fixed-bed column studies. Several mechanisms were proposed for acetaminophen removal, including dispersive interactions by π electrons, electrostatic attractions, and hydrophobic interactions. In batch mode, a removal efficiency of 97.35% was achieved for an initial acetaminophen concentration of 100 mg/L at pH 5.7, with a contact time of 24 min, a biochar dose of 15 g/L, and a temperature of 25 °C. In fixed-bed column studies, it was found that increasing the flow rate enhanced the biochar's uptake capacity, while the breakdown time decreased with variations in bed height.

Song et al. [110] investigated the use of biochar derived from *Sargassum* sp. seaweed for the biosorption of TET and cefradine (CF). The uptake of both drugs was found to be pH dependent. The maximum adsorption capacities calculated using the Langmuir model were 128.1 mg/g for TET and 61.7 mg/g for CF, with more CF molecules adsorbed at lower concentrations. The biosorption process was primarily driven by Coulombic interactions and π - π electron donor-acceptor interactions between the seaweed biochar and the TET/CF molecules.

The production of activated carbon from invasive *Sargassum* sp. biomass was optimized using response surface methodology [111]. The biosorption potential of the activated carbon was evaluated for the removal of caffeine, TET, penicillin, and methylene blue. Under optimal conditions, the calculated biosorption capacities were 329 mg/g for caffeine, 579 mg/g for TET, 150 mg/g for penicillin, and 222 mg/g for methylene blue. In a quaternary mixture with an initial concentration of 200 mg/L for each component, the observed removal rates were 39.9% for caffeine, 48.9% for TET, 39.7% for penicillin, and 70.8% for methylene blue.

Table 3 shows the maximal biosorption capacities of different algal-derived biosorbents for different pharmaceuticals.

Table 3. Algal biomass-derived biosorbents for pharmaceuticals removal.

Biosorbent	Biosorbate	Experimental Conditions	q _{max}	Ref.
<i>Pithophora</i> sp.	Amoxicillin	25 °C, pH = 5.0, C ₀ = 10–150 mg/L, W = 0.5 g/L, 3 h	25.83	[103]
<i>Posidonia oceanica</i> (chemically modified)	Oxytetracycline	25 °C, pH = 6.0, C ₀ = 50–1000 mg/L, W = 8 g/L, 1 h	16.2	[104]
<i>Sargassum cymosum</i> — biochar	Acetaminophen	25 °C, pH = 5.7, C ₀ = 9.2–80.6 mg/L, W = 3 g/L, 0.4 h	15.85	[109]
<i>Sargassum filipendula</i> (waste)	Propranolol hydrochloride	25 °C, pH = 8.5, C ₀ = 29.6–1035 mg/L, W = 2 g/L, 2 h	572.6	[105]

5. Concluding Remarks

Biosorption is an alternative and environmentally friendly method for the removal of toxicants from aqueous solutions, including pharmaceuticals, which attracts the interest of

researchers. The type of biosorbent and process parameters undoubtedly play a dominant role in the effectiveness of the process, which also explains the fact that the main part of the publications is in this direction. The efforts of the researchers are mainly directed to the structural and morphological characterization of biosorbents with the maximum number of available analytical techniques, clarification of the mechanism of biosorption, optimization of process parameters, process modeling (equilibrium, kinetic, and thermodynamic), and less frequently on the reusability of the biosorbents. At present, a wide variety of microbial and algal-derived biosorbents have been successfully studied for the removal of pharmaceuticals (mostly antibiotics and non-steroidal anti-inflammatory pharmaceuticals) from model aqueous solutions and wastewater. These are particularly important studies, but if we want the biosorption to overcome the stage of laboratory research and find a real practical application in industrial conditions, it is absolutely a must to clearly define the advantages and disadvantages of different biosorbents and to move to a stage of research in a more technological aspect.

The different types of biosorbents used for waste pharmaceuticals removal demonstrate the following advantages (A)/disadvantages (D):

Native microbial/algal biomasses—(A) low cost; (D) lower biosorption capacity, low mechanical strength, low contact surface, difficult separation of the biosorbent at the end of the process, and difficult regeneration and reuse.

Modified microbial/algal biomasses—(A) improved biosorption capacity; (D) low mechanical strength, low contact surface, need of additional technological steps for preparation of the biosorbent, difficult separation of the biosorbent at the end of the process, and difficult regeneration and reuse.

Immobilized microbial/algal biomasses—(A) improved biosorption capacity, improved mechanical strength, easier separation of the biosorbent at the end of the process, and improved opportunities for regeneration and reuse; (D) low contact surface and need for additional technological steps for preparation of the biosorbent.

Biocomposites based on microbial/algal biomasses without magnetic nanoparticles—(A) improved biosorption capacity, improved contact surface, improved mechanical strength, easier separation of the biosorbent at the end of the process, and improved opportunities for regeneration and reuse; (D) need of additional technological steps for preparation of the biosorbent.

Biocomposites based on microbial/algal biomasses with magnetic nanoparticles—(A) improved biosorption capacity, improved contact surface, improved mechanical strength, the easiest separation of the biosorbent at the end of the process by magnetic field, and improved opportunities for regeneration and reuse; (D) need for additional technological steps for preparation of the biosorbent.

According to the results analyzed, the biosorbents based on bacterial biomasses co-immobilized with magnetic nanoparticles are among the most promising materials for waste pharmaceutical removal from aqueous solutions.

Future efforts should focus on the following areas:

- (i) The use of residual microbial and algal biomass, which will contribute to the zero waste strategy.
- (ii) Construction of biocomposite biosorbents based on microbial cells co-immobilized with magnetic nanoparticles and incorporation into fixed and fluidized bed columns.
- (iii) Studies on biosorption from model wastewater solutions and real wastewater samples should receive attention, as real aqueous matrices contain organic and inorganic compounds that lead to a decrease in the removal rate of the studied toxicants.
- (iv) Pilot-scale-based biosorption research must be conducted to determine the effectiveness of pharmaceutical removal.

- (v) The combination of biosorption with another treatment process can be a promising tool for removing pharmaceuticals from wastewater.

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References

1. Sauv , S.; Desrosiers, M. A review of what is an emerging contaminant. *Chem. Cent. J.* **2014**, *8*, 15. [\[CrossRef\]](#)
2. Feng, W.; Deng, Y.; Yang, F.; Miao, Q.; Ngien, S.K. Systematic Review of Contaminants of Emerging Concern (CECs): Distribution, Risks, and Implications for Water Quality and Health. *Water* **2023**, *15*, 3922. [\[CrossRef\]](#)
3. Pastorino, P.; Ginebreda, A. Contaminants of Emerging Concern (CECs): Occurrence and Fate in Aquatic Ecosystems. *Int. J. Environ. Res. Public Health* **2021**, *18*, 13401. [\[CrossRef\]](#)
4. Osuoha, J.O.; Anyanwu, B.O.; Ejileugha, C. Pharmaceuticals and personal care products as emerging contaminants: Need for combined treatment strategy. *J. Hazard. Mater. Adv.* **2023**, *9*, 100206. [\[CrossRef\]](#)
5. Morin-Crini, N.; Lichtfouse, E.; Liu, G.; Balaram, V.; Ribeiro, A.R.L.; Lu, Z.; Stock, F.; Carmona, E.; Teixeira, M.R.; Picos-Corrales, L.A.; et al. Worldwide cases of water pollution by emerging contaminants: A review. *Environ. Chem. Lett.* **2022**, *20*, 2311–2338. [\[CrossRef\]](#)
6. Praveena, S.M.; Cheema, M.S.; Guo, H.R. Non-nutritive artificial sweeteners as an emerging contaminant in environment: A global review and risks perspectives. *Ecotoxicol. Environ. Saf.* **2019**, *170*, 669–707. [\[CrossRef\]](#) [\[PubMed\]](#)
7. Pastorino, P.; Pizzul, E.; Bertoli, M.; Anselmi, S.; Ku ce, M.; Menconi, V.; Prearo, M.; Renzi, M. First insights into plastic and microplastic occurrence in biotic and abiotic compartments, and snow from a high-mountain lake (Carnic Alps). *Chemosphere* **2020**, *265*, 129121. [\[CrossRef\]](#) [\[PubMed\]](#)
8. Sicuro, B.; Pastorino, P.; Barbero, R.; Barisone, S.; Dellerba, D.; Menconi, V.; Righetti, M.; De Vita, V.; Prearo, M. Prevalence and antibiotic sensitivity of bacteria isolated from imported ornamental fish in Italy: A translocation of resistant strains? *Prev. Vet. Med.* **2020**, *175*, 104880. [\[CrossRef\]](#) [\[PubMed\]](#)
9. Pic , Y.; Barcel , D. Mass Spectrometry in Wastewater-Based Epidemiology for the Determination of Small and Large Molecules as Biomarkers of Exposure: Toward a Global View of Environment and Human Health under the COVID-19 Outbreak. *ACS Omega* **2021**, *6*, 30865–30872. [\[CrossRef\]](#) [\[PubMed\]](#)
10. Daughton, C.G. Non-regulated water contaminants: Emerging research. *Environ. Impact Assess. Rev.* **2004**, *24*, 711–732. [\[CrossRef\]](#)
11. Tabak, H.H.; Bunch, R.L. Steroid hormones as water pollutants I. Metabolism of natural and synthetic ovulation-inhibiting hormones by microorganisms of activated sludge primary settled sewage. *Dev. Ind. Microbiol.* **1970**, *11*, 367–376.
12. Norpoth, K.; Nehrkorn, A.; Kirchner, M.; Holsen, H.; Teipel, H. Investigations on the problem of solubility and stability of steroid ovulation inhibitors in water, wastewater and activated sludge. *Zentralbl. Bakteriол. Parasitenkd. Infekt. Hyg. Erste Abt. Orig.* **1973**, *156*, 500–511.
13. K mmerer, K. (Ed.) Introduction: Pharmaceuticals in the Environment. In *Pharmaceuticals in the Environment*, 1st ed.; Springer: Berlin/Heidelberg, Germany, 2001; pp. 67–114.
14. K mmerer, K. The presence of pharmaceuticals in the environment due to human use—present knowledge and future challenges. *J. Environ. Manag.* **2009**, *90*, 2354–2366. [\[CrossRef\]](#) [\[PubMed\]](#)
15. Tannoury, M.; Attieh, Z. The Influence of Emerging Markets on the Pharmaceutical Industry. *Curr. Ther. Res. Clin. Exp.* **2017**, *86*, 19–22. [\[CrossRef\]](#)
16. Gonz lez Pe a, O.I.; L pez Zavala, M. .; Cabral Ruelas, H. Pharmaceuticals Market, Consumption Trends and Disease Incidence Are Not Driving the Pharmaceutical Research on Water and Wastewater. *Int. J. Environ. Res. Public Health* **2021**, *18*, 2532. [\[CrossRef\]](#)
17. Aidara-Kane, A.; Angulo, F.J.; Conly, J.M.; Minato, Y.; Silbergeld, E.K.; McEwen, S.A.; Collignon, P.J. World Health Organization (WHO) guidelines on use of medically important antimicrobials in food-producing animals. *Antimicrob. Resist. Infect. Control* **2018**, *7*, 7. [\[CrossRef\]](#) [\[PubMed\]](#)
18. Gworek, B.; Kij nska, M.; Wrzosek, J.; Graniewska, M. Pharmaceuticals in the Soil and Plant Environment: A Review. *Water Air Soil Pollut.* **2021**, *232*, 145. [\[CrossRef\]](#)

19. Wang, J.; Chu, L.; Wojnárovits, L.; Takács, E. Occurrence and fate of antibiotics, antibiotic resistant genes (ARGs) and antibiotic resistant bacteria (ARB) in municipal wastewater treatment plant: An overview. *Sci. Total Environ.* **2020**, *744*, 140997. [\[CrossRef\]](#)
20. Pereira, A.; Silva, L.; Laranjeiro, C.; Pena, A. Assessment of Human Pharmaceuticals in Drinking Water Catchments, Tap and Drinking Fountain. *Waters Appl. Sci.* **2021**, *11*, 7062. [\[CrossRef\]](#)
21. Ma, L.; Liu, Y.; Yang, Q.; Jiang, L.; Li, G. Occurrence and distribution of Pharmaceuticals and Personal Care Products (PPCPs) in wastewater related riverbank groundwater. *Sci. Total Environ.* **2022**, *821*, 153372. [\[CrossRef\]](#) [\[PubMed\]](#)
22. Couto, C.F.; Lange, L.C.; Amaral, M.C.S. Occurrence, fate and removal of pharmaceutically active compounds (PhACs) in water and wastewater treatment plants-A review. *J. Water Process Eng.* **2019**, *32*, 100927. [\[CrossRef\]](#)
23. OECD (Organization for Economic Co-operation and Development). *Pharmaceutical Residues in Freshwater: Hazards and Policy Responses, OECD Studies on Water*; OECD Publishing: Paris, France, 2019.
24. Kidd, A.; Blanchfield, P.; Mills, K.; Palace, V.; Evans, R.; Lazorchak, J.; Flick, R. Collapse of a fish population after exposure to synthetic estrogen. *Proc. Natl. Acad. Sci. USA* **2007**, *104*, 8897–8901. [\[CrossRef\]](#) [\[PubMed\]](#)
25. Gould, S.L.; Winter, M.; Norton, W.; Tyler, C. The potential for adverse effects in fish exposed to antidepressants in the aquatic environment. *Environ. Sci. Technol.* **2021**, *55*, 16299–16312. [\[CrossRef\]](#) [\[PubMed\]](#)
26. Quinn, B.; Gagné, F.; Blaise, C. Evaluation of the acute, chronic and teratogenic effects of a mixture of eleven pharmaceuticals on the cnidarian, *Hydra attenuata*. *Sci. Total Environ.* **2009**, *407*, 1072–1079. [\[CrossRef\]](#) [\[PubMed\]](#)
27. Verdú, J.; Cortez, V.; Ortiz, A.; González-Rodríguez, E. Low doses of ivermectin cause sensory and locomotor disorders in dung beetles. *Sci. Rep.* **2015**, *5*, 13912. [\[CrossRef\]](#)
28. Reyes, N.J.D.G.; Geronimo, F.K.F.; Yano, K.A.V.; Guerra, H.B.; Kim, L.-H. Pharmaceutical and personal care products in different matrices: Occurrence, pathways, and treatment processes. *Water* **2021**, *13*, 1159. [\[CrossRef\]](#)
29. Rout, P.R.; Zhang, T.C.; Bhunia, P.; Surampalli, R.Y. Treatment technologies for emerging contaminants in wastewater treatment plants: A review. *Sci. Total Environ.* **2021**, *753*, 141990. [\[CrossRef\]](#)
30. Anumol, T.; Vijayanandan, A.; Park, M.; Philip, L.; Snyder, S.A. Occurrence and fate of emerging trace organic chemicals in wastewater plants in Chennai, India. *Environ. Int.* **2016**, *92*, 33–42. [\[CrossRef\]](#)
31. Bueno, M.M.; Gomez, M.J.; Herrera, S.; Hernando, M.D.; Agüera, A.; Fernández-Alba, A.R. Occurrence and persistence of organic emerging contaminants and priority pollutants in five sewage treatment plants of Spain: Two years pilot survey monitoring. *Environ. Pollut.* **2012**, *164*, 267–273. [\[CrossRef\]](#)
32. Comber, S.; Gardner, M.; Sörme, P.; Leverett, D.; Ellor, B. Active pharmaceutical ingredients entering the aquatic environment from wastewater treatment works: A cause for concern? *Sci. Total Environ.* **2018**, *613*, 538–547. [\[CrossRef\]](#) [\[PubMed\]](#)
33. Ahmed, M.B.; Zhou, J.L.; Ngo, H.H.; Guo, W.; Thomaidis, N.S.; Xu, J. Progress in the biological and chemical treatment technologies for emerging contaminant removal from wastewater: A critical review. *J. Hazard. Mater.* **2017**, *323 Pt A*, 274–298. [\[CrossRef\]](#)
34. Stadlmair, L.F.; Letzel, T.; Drewes, J.E.; Grassman, J. Enzymes in removal of pharmaceuticals from wastewater: A critical review of challenges, applications and screening methods for their selection. *Chemosphere* **2018**, *205*, 649–661. [\[CrossRef\]](#)
35. Kumar, V.; Lakkaboyana, S.K.; Sharma, N.; Chakraborty, P.; Umesh, M.; Pasrija, R.; Thomas, J.; Kalebar, V.U.; Jayaraj, I.; Awasthi, M.K.; et al. A critical assessment of technical advances in pharmaceutical removal from wastewater—A critical review. *CSCEE* **2023**, *8*, 100363. [\[CrossRef\]](#)
36. Ghazal, H.; Koumaki, E.; Hoslett, J.; Malamis, S.; Katsou, E.; Barcelo, D.; Jouhara, H. Insights into current physical, chemical and hybrid technologies used for the treatment of wastewater contaminated with pharmaceuticals. *J. Clean. Prod.* **2022**, *361*, 132079. [\[CrossRef\]](#)
37. Ilavský, J.; Barloková, D. The Removal of Selected Pharmaceuticals from Water by Adsorption with Granular Activated Carbons. *Eng. Proc.* **2023**, *57*, 33. [\[CrossRef\]](#)
38. Gadd, G.M. Biosorption: Critical Review of Scientific Rationale, Environmental Importance and Significance for Pollution Treatment. *J. Chem. Technol. Biotechnol.* **2009**, *84*, 13–28. [\[CrossRef\]](#)
39. Salam, K.A. Towards sustainable development of microalgal biosorption for treating effluents containing heavy metals. *Biofuel Res. J.* **2019**, *22*, 948–961. [\[CrossRef\]](#)
40. Velkova, Z.; Kirova, G.; Gochev, V.; Hristova, Y.; Zlatev, R.; Stoytcheva, M. Biomass-Derived biosorbents for toxic metals removal. In *Biomass: The Novel Green Gold: Current Trends and Future Uses of Biomass Resources*, 1st ed.; Ortega, M., Bautista, J., Eds.; Nova Science Publishers: New York, NY, USA, 2024; Chapter 4; pp. 131–180.
41. Modak, J.M.; Natarajan, K.A. Biosorption of metals using nonliving biomass—A review. *Min. Metall. Explor.* **1995**, *12*, 189–196. [\[CrossRef\]](#)
42. Vijayaraghavan, K.; Yun, Y.S. Bacterial biosorbents and biosorption. *Biotechnol. Adv.* **2008**, *26*, 266–291. [\[CrossRef\]](#) [\[PubMed\]](#)
43. Velkova, Z.; Kirova, G.; Stoytcheva, M.; Kostadinova, S.; Todorova, K.; Gochev, V. Immobilized microbial biosorbents for heavy metals removal. *Eng. Life Sci.* **2018**, *18*, 871–881. [\[CrossRef\]](#) [\[PubMed\]](#)

44. Thirunavukkarasu, A.; Nithya, R.; Sivashankar, R. Continuous fixed-bed biosorption process: A review. *J. Adv. Chem. Eng.* **2021**, *8*, 100188. [\[CrossRef\]](#)
45. Tofan, L.; Suteu, D. Renewable Resource Biosorbents for Pollutant Removal from Aqueous Effluents in Column Mode. *Separations* **2023**, *10*, 143. [\[CrossRef\]](#)
46. Torres, E. Biosorption: A Review of the Latest Advances. *Processes* **2020**, *8*, 1584. [\[CrossRef\]](#)
47. Michalak, I.; Chojnacka, K.; Witek-Krowiak, A. State of the Art for the Biosorption Process—a Review. *Appl. Biochem. Biotechnol.* **2013**, *170*, 1389–1416. [\[CrossRef\]](#) [\[PubMed\]](#)
48. Rusu, L.; Suceveanu, E.-M.; Blaga, A.-C.; Nedeff, F.M.; Şuteu, D. Insights into Recent Advances of Biomaterials Based on Microbial Biomass and Natural Polymers for Sustainable Removal of Pharmaceuticals Residues. *Polymers* **2023**, *15*, 2923. [\[CrossRef\]](#) [\[PubMed\]](#)
49. Pellenz, L.; Silva de Oliveira, C.R.; da Silva Júnior, A.H.; Souza da Silva, L.J.; da Silva, L.; de Souza, A.A.U.; de Arruda Guelli Ulson de Souza, S.M.; Borba, F.H. A comprehensive guide for characterization of adsorbent materials. *Sep. Purif. Technol.* **2023**, *305*, 122435. [\[CrossRef\]](#)
50. Chojnacka, K.; Mikulewicz, M. Green analytical methods of metals determination in biosorption studies. *TrAC Trends Anal. Chem.* **2019**, *116*, 254–265. [\[CrossRef\]](#)
51. Bush, K. Antimicrobial agents. *Curr. Opin. Chem. Biol.* **1997**, *1*, 169–175. [\[CrossRef\]](#) [\[PubMed\]](#)
52. Aryal, M. A comprehensive study on the bacterial biosorption of heavy metals: Materials, performances, mechanisms, and mathematical modellings. *Rev. Chem. Eng.* **2021**, *37*, 715–754. [\[CrossRef\]](#)
53. Van Hullebusch, E.D.; Zandvoort, M.H.; Lens, P.N. Metal immobilization by biofilms: Mechanisms and analytical tools. *Rev. Environ. Sci. Biotechnol.* **2003**, *2*, 9–33. [\[CrossRef\]](#)
54. Ahluwalia, S.S.; Goyal, D. Microbial and plant derived biomass for removal of heavy metals from wastewater. *Bioresour. Technol.* **2007**, *98*, 2243–2257. [\[CrossRef\]](#) [\[PubMed\]](#)
55. Pham, V.H.T.; Kim, J.; Chang, S.; Chung, W. Bacterial Biosorbents, an Efficient Heavy Metals Green Clean-Up Strategy: Prospects, Challenges, and Opportunities. *Microorganisms* **2022**, *10*, 610. [\[CrossRef\]](#)
56. Al-Gheethi, A.A.; Efaa, A.N.; Mohamed, R.M.; Norli, I.; Kadir, M.O. Potential of bacterial consortium for removal of cephalexin from aqueous solution. *J. Assoc. Arab Univ. Basic Appl. Sci.* **2017**, *24*, 141–188. [\[CrossRef\]](#)
57. Kirova, G.; Velkova, Z.; Stoytcheva, M.; Gochev, V. Tetracycline removal from model aqueous solutions by pretreated waste *Streptomyces fradiae* biomass. *Biotechnol. Biotechnol. Equip.* **2021**, *35*, 953–963. [\[CrossRef\]](#)
58. Ertas, E.; Tural, B.; Tural, S. Biosorption of diclofenac by magnetic biosorbent (M-*E.Coli*). *Iran. J. Chem. Chem. Eng.* **2024**, *43*, 1482–1494.
59. Gopal, G.; Roy, N.; Chandrasekaran, N.; Mukherjee, A. Photo-Assisted Removal of Tetracycline Using Bio-Nanocomposite-Immobilized Alginate Beads. *ACS Omega* **2019**, *4*, 17504–17510. [\[CrossRef\]](#)
60. Niță, N.-T.; Suceveanu, E.-M.; Nedeff, F.M.; Tița, O.; Rusu, L. Biocomposite Material Based on *Lactococcus lactis* sp. Immobilized in Natural Polymer Matrix for Pharmaceutical Removal from Aqueous Media. *Polymers* **2024**, *16*, 1804. [\[CrossRef\]](#) [\[PubMed\]](#)
61. Pathak, J.; Rajneesh; Maurya, P.K.; Singh, S.P.; Haeder, D.P.; Sinha, R.P. Cyanobacterial Farming for Environment Friendly Sustainable Agriculture Practices: Innovations and Perspectives. *Front. Environ. Sci.* **2018**, *6*, 7. [\[CrossRef\]](#)
62. Momin, S.C.; Nath, J.; Mahana, A.; Pradhan, R.B.; Mehta, S.K. Chemical activation of *Microcystis aeruginosa* biomass: A promising approach for enhanced diclofenac sorption and water treatment. *J. Chem. Technol. Biotechnol.* **2024**, *99*, 149–163. [\[CrossRef\]](#)
63. Santaefemia, S.; Torres, E.; Abalde, J. Biosorption of ibuprofen from aqueous solution using living and dead biomass of the microalga *Phaeodactylum tricornutum*. *J. Appl. Phycol.* **2018**, *30*, 471–482. [\[CrossRef\]](#)
64. Mem, A.; Abd El-Aty, A.M.; Badawy, M.I.; Ali, R.K. Removal of pharmaceutical pollutants from synthetic wastewater using chemically modified biomass of green alga *Scenedesmus obliquus*. *Ecotoxicol. Environ. Saf.* **2018**, *151*, 144–152.
65. Coimbra, R.N.; Escapa, C.; Vázquez, N.C.; Noriega-Hevia, G.; Otero, M. Utilization of Non-Living Microalgae Biomass from Two Different Strains for Adsorptive Removal of Diclofenac from Water. *Water* **2018**, *10*, 1401. [\[CrossRef\]](#)
66. Silva, A.; Coimbra, R.N.; Escapa, C.; Figueiredo, S.A.; Freitas, O.M.; Otero, M. Green Microalgae *Scenedesmus obliquus* Utilization for the Adsorptive Removal of Nonsteroidal Anti-Inflammatory Drugs (NSAIDs) from Water Samples. *Int. J. Environ. Res. Public Health* **2020**, *17*, 3707. [\[CrossRef\]](#) [\[PubMed\]](#)
67. Zambrano, J.; García-Encina, P.A.; Hernández, F.; Botero-Coy, A.M.; Jiménez, J.J.; Irusta-Mata, R. Removal of a mixture of veterinary medicinal products by adsorption onto a *Scenedesmus almeriensis* microalgae-bacteria consortium. *J. Water Process Eng.* **2021**, *43*, 102226. [\[CrossRef\]](#)
68. Habibzadeh, M.; Chaibakhsh, N.; Naemi, A.S. Optimized treatment of wastewater containing cytotoxic drugs by living and dead biomass of the freshwater microalga, *Chlorella vulgaris*. *Ecol. Eng.* **2018**, *111*, 85–93. [\[CrossRef\]](#)
69. Mirizadeh, S.; Solisio, C.; Converti, A.; Casazza, A.A. Efficient removal of tetracycline, ciprofloxacin, and amoxicillin by novel magnetic chitosan/microalgae biocomposites. *Sep. Purif. Technol.* **2024**, *329*, 125115. [\[CrossRef\]](#)

70. Esmaili, Z.; Barikbin, B.; Shams, M.; Alidadi, H.; Al-Musawi, T.J.; Bonyadi, Z. Biosorption of metronidazole using *Spirulina platensis* microalgae: Process modeling, kinetic, thermodynamic, and isotherm studies. *Appl. Water Sci.* **2023**, *13*, 63. [\[CrossRef\]](#)
71. Escapa, C.; Coimbra, R.N.; Nuevo, C.; Vega, S.; Paniagua, S.; García, A.I.; Calvo, L.F.; Otero, M. Valorization of Microalgae Biomass by Its Use for the Removal of Paracetamol from Contaminated Water. *Water* **2017**, *9*, 312. [\[CrossRef\]](#)
72. Suárez-Martínez, D.; Angulo-Mercado, E.; Mercado-Martínez, I.; Vacca-Jimeno, V.; Tapia-Larios, C.; Cubillán, N. Enhanced Tetracycline Removal from Highly Concentrated Aqueous Media by Lipid-Free *Chlorella* sp. Biomass. *ACS Omega* **2022**, *7*, 14128–14137. [\[CrossRef\]](#) [\[PubMed\]](#)
73. Saldaña, K.; Angulo, E.; Mercado, I.; Castellar, G.; Cubillán, N. Removal of minocycline from high concentrated aqueous medium by nonliving and lipid-free *Chlorella* sp. biomass. *Bioresour. Technol.* **2022**, *17*, 100921. [\[CrossRef\]](#)
74. Garcia-Rubio, R.; de Oliveira, H.C.; Rivera, J.; Trevijano-Contador, N. The Fungal Cell Wall: *Candida*, *Cryptococcus*, and *Aspergillus* Species. *Front. Microbiol.* **2020**, *10*, 2993. [\[CrossRef\]](#)
75. Dhankhar, R.; Hooda, A. Fungal biosorption—an alternative to meet the challenges of heavy metal pollution in aqueous solutions. *Environ. Technol.* **2011**, *32*, 467–491. [\[CrossRef\]](#)
76. Wang, J.L.; Chen, C. Biosorption of heavy metals by *Saccharomyces cerevisiae*: A review. *Biotechnol. Adv.* **2006**, *24*, 427–451. [\[CrossRef\]](#)
77. Teparić, R.; Lozančić, M.; Mrša, V. Evolutionary Overview of Molecular Interactions and Enzymatic Activities in the Yeast Cell Walls. *Int. J. Mol. Sci.* **2020**, *21*, 8996. [\[CrossRef\]](#) [\[PubMed\]](#)
78. Samarghandi, M.R.; Asgari, G.; Shokoohi, R.; Dargahic, A.; Arabkouhsar, A. Removing amoxicillin antibiotic from aqueous solutions by *Saccharomyces cerevisiae* yeast bioadsorbent: Kinetic, thermodynamic and isotherm studies. *Desalin. Water Treat.* **2019**, *152*, 306–315. [\[CrossRef\]](#)
79. Santos, B.A.P.; Dall'Oglio, E.L.; de Siqueira, S.; Caixeta, D.S.; Padilha Lopes, V.-C.; de Vasconcelos, L.; Beraldo de Moraes, E. Ibuprofen biosorption by chemically activated *Saccharomyces cerevisiae*. *Rev. Ambient. Água* **2022**, *17*, e2862. [\[CrossRef\]](#)
80. Debs, K.B.; da Silva, H.D.T.; de Lourdes Leite de Moraes, M.; Carrilho, E.N.V.M.; Lemos, S.G.; Labuto, G. Biosorption of 17 α -ethynylestradiol by yeast biomass from ethanol industry in the presence of estrone. *Environ. Sci. Pollut. Res.* **2019**, *26*, 28419–28428. [\[CrossRef\]](#) [\[PubMed\]](#)
81. Rusu, L.; Grigoraş, C.-G.; Suceveanu, E.M.; Simion, A.-I.; Dediu Botezatu, A.V.; Istrate, B.; Doroftei, I. Eco-Friendly Biosorbents Based on Microbial Biomass and Natural Polymers: Synthesis, Characterization and Application for the Removal of Drugs and Dyes from Aqueous Solutions. *Materials* **2021**, *14*, 4810. [\[CrossRef\]](#)
82. Rusu, L.; Grigoraş, C.-G.; Simion, A.-I.; Suceveanu, E.M.; Şuteu, D.; Harja, M. Application of *Saccharomyces cerevisiae*/Calcium Alginate Composite Beads for Cephalexin Antibiotic Biosorption from Aqueous Solutions. *Materials* **2021**, *14*, 4728. [\[CrossRef\]](#)
83. Rusu, L.; Grigoraş, C.-G.; Simion, A.-I.; Suceveanu, E.-M.; Istrate, B.; Harja, M. Biosorption Potential of Microbial and Residual Biomass of *Saccharomyces pastorianus* Immobilized in Calcium Alginate Matrix for Pharmaceuticals Removal from Aqueous Solutions. *Polymers* **2022**, *14*, 2855. [\[CrossRef\]](#) [\[PubMed\]](#)
84. Rusu, L.; Grigoraş, C.-G.; Simion, A.-I.; Suceveanu, E.-M.; Dediu Botezatu, A.V.; Harja, M. Biosorptive Removal of Ethacridine Lactate from Aqueous Solutions by *Saccharomyces pastorianus* Residual Biomass/Calcium Alginate Composite Beads: Fixed-Bed Column Study. *Materials* **2022**, *15*, 4657. [\[CrossRef\]](#)
85. Kip, F.; Açikel, Ü. Removal of tetracycline by biocomposites synthesized with immobilization of *Rhizopus delamar* and *Candida* types. *J. Fac. Eng. Archit. Gazi Univ.* **2019**, *34*, 1417–1426.
86. Pokethitiyook, P.; Poolpak, T. Biosorption of heavy metal from aqueous solutions. In *Phytoremediation: Management of Environmental Contaminants*, 1st ed.; Ansari, A.A., Gill, S.S., Gill, R., Lanza, G.R., Newman, L., Eds.; Springer: Cham, Switzerland, 2016; Volume 3, Chapter 4; pp. 113–141.
87. Aksu, Z.; Tunç, Ö. Application of biosorption for penicillin G removal: Comparison with activated carbon. *Process. Biochem.* **2005**, *40*, 831–847. [\[CrossRef\]](#)
88. Melgoza, B.; León-Santisteban, H.; López-Medina, R.; Tomasini, A. Naproxen Sorption by Non-viable *Rhizopus oryzae* Biomass. *Water Air Soil Pollut.* **2020**, *231*, 30. [\[CrossRef\]](#)
89. Azamateslamtalab, E.; Madani, M.; Ramavandi, B.; Mohammadi, R. Sonication alkaline-assisted preparation of *Rhizopus oryzae* biomass for facile bio-elimination of tetracycline antibiotic from an aqueous matrix. *Environ. Sci. Pollut. Res.* **2020**, *27*, 35792–35801. [\[CrossRef\]](#)
90. Azamateslamtalab, E.; Madani, M.; Ramavandi, B.; Mohammadi, R. Fe₃O₄-magnetized *Rhizopus oryzae* and *Penicillium citrinum* biomass for elimination of tetracycline antibiotic. *Biomass Convers. Biorefin.* **2024**, *14*, 29673. [\[CrossRef\]](#)
91. Kori, A.K.; Ramavandi, B.; Mahmoodi, S.M.M.; Javanmardi, F. Magnetization and ZIF-67 modification of *Aspergillus flavus* biomass for tetracycline removal from aqueous solutions: A stable and efficient composite. *Environ. Res.* **2024**, *252* pt 2, 118931.
92. Jureczko, M.; Przysaś, W. Removal of two cytostatic drugs: Bleomycin and vincristine by white-rot fungi—A sorption study. *J. Environ. Health Sci. Engineer.* **2021**, *19*, 651–662. [\[CrossRef\]](#)

93. Jureczko, M.; Przysaś, W. Ecotoxicity risk of presence of two cytostatic drugs: Bleomycin and vincristine and their binary mixture in aquatic environment. *Ecotoxicol. Environ. Saf.* **2019**, *172*, 210–215. [\[CrossRef\]](#) [\[PubMed\]](#)
94. Haroune, L.; Saibi, S.; Bellenger, J.P.; Cabana, H. Evaluation of the efficiency of *Trametes hirsuta* for the removal of multiple pharmaceutical compounds under low concentrations relevant to the environment. *Bioresour. Technol.* **2014**, *171*, 199–202. [\[CrossRef\]](#)
95. Lucas, D.; Castellet-Rovira, F.; Villagrasa, M.; Badia-Fabregat, M.; Barceló, D.; Vicent, T.; Caminal, G.; Sarrà, M.; Rodríguez-Mozaz, S. The role of sorption processes in the removal of pharmaceuticals by fungal treatment of wastewater. *Sci. Total Environ.* **2018**, *610*, 1147–1153. [\[CrossRef\]](#) [\[PubMed\]](#)
96. Nguyen, L.N.; Hai, F.I.; Yang, S.; Kang, J.; Leusch, F.D.L.; Roddick, F.; Price, W.E.; Nghiem, L.D. Removal of pharmaceuticals, steroid hormones, phytoestrogens, UV-filters, industrial chemicals and pesticides by *Trametes versicolor*: Role of biosorption and biodegradation. *Int. Biodeterior. Biodegrad.* **2014**, *88*, 169–175. [\[CrossRef\]](#)
97. Menk, J.J.; do Nascimento, A.I.S.; Leite, F.G.; de Oliveira, R.A.; Jozala, A.F.; de Oliveira Junior, J.M.; Chaud, M.V.; Grotto, D. Biosorption of pharmaceutical products by mushroom stem waste. *Chemosphere* **2019**, *237*, 124515. [\[CrossRef\]](#)
98. Sarikaya, G.; Osman, A.; Ozer, T.E. Biosorption of tetracycline antibiotics by *Lactarius deliciosus* biomass. *Chem. Eng. Commun.* **2024**, *211*, 592–602. [\[CrossRef\]](#)
99. Romera, E.; González, F.; Ballester, A.; Blázquez, M.L.; Muñoz, J.A. Comparative Study of Biosorption of Heavy Metals Using Different Types of Algae. *Bioresour. Technol.* **2007**, *98*, 3344–3353. [\[CrossRef\]](#)
100. Raize, O.; Argaman, Y.; Yannai, S. Mechanisms of Biosorption of Different Heavy Metals by Brown Marine Macroalgae. *Biotechnol. Bioeng.* **2004**, *87*, 451–458. [\[CrossRef\]](#) [\[PubMed\]](#)
101. Navarro, A.E.; Lim, H.; Chang, E.; Lee, Y.; Manrique, A.S. Uptake of Sulfa Drugs from Aqueous Solutions by Marine Algae. *Sep. Sci. Technol.* **2014**, *49*, 2175–2181. [\[CrossRef\]](#)
102. Khamayseh, M.M.; Kidak, R. Equilibrium, kinetics, and thermodynamics study on the biosorption of reactive levofloxacin antibiotic on *Pithophora* macroalgae in aqueous solution. *Environ. Monit. Assess.* **2023**, *19*, 301. [\[CrossRef\]](#) [\[PubMed\]](#)
103. Khamayseh, M.M.; Kidak, R. Biosorption of reactive amoxicillin antibiotic on *Pithophora* macroalgae in aqueous solution: Equilibrium and kinetic studies. *Desalin. Water Treat.* **2024**, *320*, 100669. [\[CrossRef\]](#)
104. Ferchichi, K.; Amdouni, N.; Chevalier, Y.; Hbaieb, S. Low-cost *Posidonia oceanica* bio-adsorbent for efficient removal of antibiotic oxytetracycline from water. *Environ. Sci. Pollut. Res.* **2022**, *29*, 83112–83125. [\[CrossRef\]](#) [\[PubMed\]](#)
105. Coelho, C.M.; de Andrade, J.R.; da Silva, M.G.C.; Vieira, M.G.A. Removal of propranolol hydrochloride by batch biosorption using remaining biomass of alginate extraction from *Sargassum filipendula* algae. *Environ. Sci. Pollut. Res. Int.* **2020**, *27*, 16599–16611. [\[CrossRef\]](#)
106. López-Miranda, J.L.; Molina, G.A.; Esparza, R.; González-Reyna, A.M.; Silva, R.; Estévez, M.S.R. Ecofriendly and sustainable *Sargassum* spp.-based system for the removal of highly used drugs during the COVID-19 pandemic. *Arab. J. Chem.* **2022**, *15*, 104169. [\[CrossRef\]](#) [\[PubMed\]](#)
107. Tan, X.; Zhang, F.; Wang, H.; Ho, S.-H. The magic of algae-based biochar: Advantages, preparation, and applications. *Bioengineered* **2023**, *14*, 2252157. [\[CrossRef\]](#) [\[PubMed\]](#)
108. Patel, A.K.; Katiyar, R.; Chen, C.W.; Singhanian, R.R.; Awasthi, M.K.; Bhatia, S.; Bhaskar, T.; Dong, C.D. Antibiotic bioremediation by new generation biochar: Recent updates. *Bioresour. Technol.* **2022**, *358*, 127384. [\[CrossRef\]](#) [\[PubMed\]](#)
109. Pimentel-Almeida, W.; Itokazu, A.G.; Bazani, H.A.G.; Maraschin, M.; Rodrigues, O.H.C.; Corrêa, R.G.; Lopes, S.; Almerindo, G.I.; Moresco, R. Beach-cast *Sargassum cymosum* macroalgae: Biochar production and apply to adsorption of Acetaminophen in batch and fixed-bed adsorption processes. *Environ. Technol.* **2023**, *44*, 974–987. [\[CrossRef\]](#) [\[PubMed\]](#)
110. Song, G.; Guo, J.; Li, G.; Zhao, W.; Yu, Y. Comparison for adsorption of tetracycline and cefradine using biochar derived from seaweed *Sargassum* sp. *Desalin. Water Treat.* **2019**, *160*, 316–324. [\[CrossRef\]](#)
111. Francoeur, M.; Yacou, C.; Jean-Marius, C.; Chérémont, Y.; Jauregui-Haza, U.; Gaspard, S. Optimization of the synthesis of activated carbon prepared from *Sargassum* (sp.) and its use for tetracycline, penicillin, caffeine and methylene blue adsorption from contaminated water. *Environ. Technol. Innov.* **2022**, *28*, 102940. [\[CrossRef\]](#)

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